

CS324 Graphics

Instructions:

To run the game, first download the file submitted for the coursework and extract the zip. Then once extracted, run a python localhost server. Open the server on your browser. Then navigate to the extracted file 'Coursework', and open the html file cw.html. From there the first scene will begin to instantly load. This may take a bit of time, but you will be able to start up the game from there! Note: when level 2 starts, if you cannot click to start it, press escape, and then click again and it will begin.

Introduction

This game, Harold's Happy House, is one very similar to the many horror games that dominated the 2010s. Inspired by the likes of Five Nights at Freddy's, and especially Petscop, this game has you take control of a small character as they explore a soft play emulation. The game's introduction is designed to set up existing lore and story behind the game's existence, to be more than just this coursework - however everything depicted is purely fiction. This style was chosen for a chance to work with how lighting can massively change and impact 3D environments, with lots of light giving a warm comfort, and then the sudden absence of it can unnerve the player significantly.

Controls

The controls were a large undertaking in this game. I knew I wanted it to be first person, so that I don't need to create a render and animations for the playable character, however this brought about its own issues. The first person camera was initially designed by hand, like what was used in simondevyoutube tutorial, however this capped off when turning in each direction. What I wanted to use was Pointerlock. While the in-built addon was used at first, it then was shifted to more closely resemble the schteppe example. This was done as it gave me much more control over the details and attributes of the camera. The solution was rewritten and wrapped in an appropriate class, which then was adapted to work with the physics.

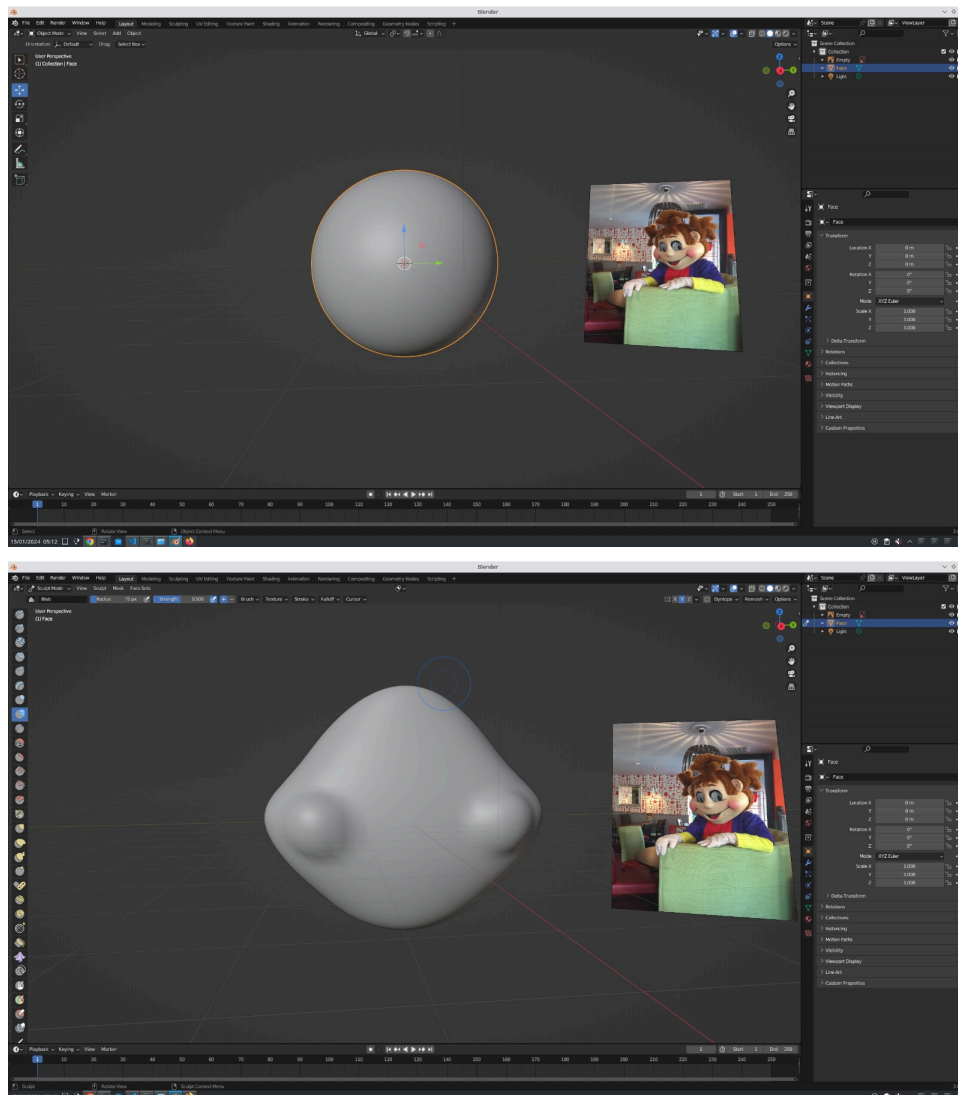
Physics

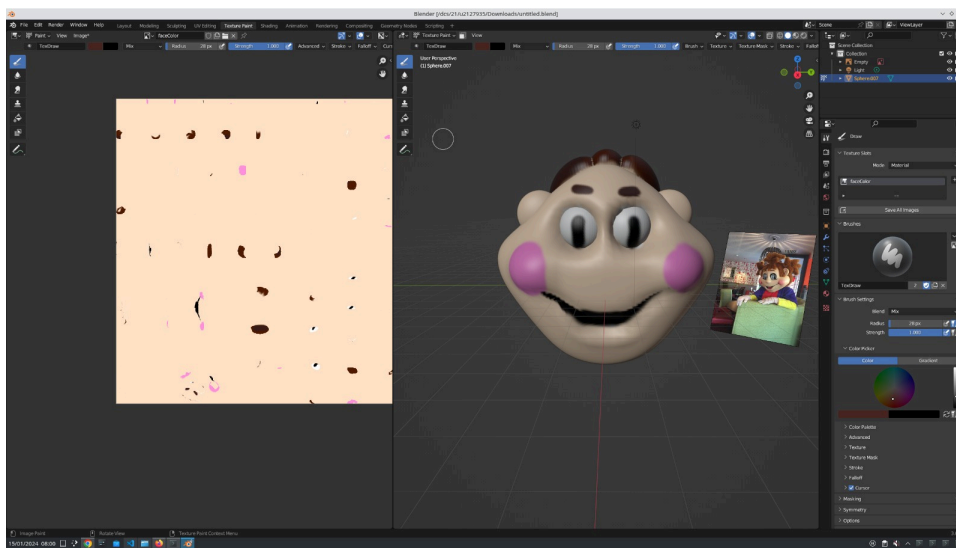
The physics engine used was Cannon.js, which was chosen as it was more lightweight than Ammo.js, and the physics required were not that complex. The physics were built with the principle that there is a physics world and a graphical world, where the physics world exclusively influences the graphics world. For static objects of no mass, this meant very little, but for moving objects and the player, on animate the mesh was synced to the physics body in position and

rotation. The player provided a difficult challenge, as they received external inputs which impacted them. The solution decided upon was the one used by the pmndrs example, however because slidey gameplay is my ultimate pet peeve, it was reworked for appropriate linear damping across the x and z axis.

Modeling

As per the description of the task, there was a model created for the game. While I knew I would want a torch in the game, and that would have been easier to make, I really wanted to create something that emulated the creepy energy of 2010 style games, and with my chosen lore and setting I knew what I wanted to make. I took inspiration from the Wacky Warehouse mascot, as shown in the images as the reference. Due to scope, I limited myself to only making the head of the mascot over the body. I first started with a sphere, which I sculpted using the grab tool and inflate tool to make it a very angular shape with large cheeks.





Proceeding this, I began to use the inverse draw tool, and smooth tool to chip away to give two large eye holes and a smiling mouth. The model then was given ears and eyes, in the form of augmented spheres. As for the hair, I stretched out several spheres into long cylinder-like tubes and pulled them back like it was slipped back hair. This wasn't exactly to the reference, but simplified to avoid the difficulties of that mascot's crazy hair. Once the face shape was sculpted, I unwrapped the UVs and began to texture map. I applied the flesh tone to the whole shape and added the key colors of the hair, the mouth, and the cheeks. After all of this, I added a cute nose as I forgot to in the images.

Assets

The assets used for this project come from a large variety of places. As the setting of this game was to take place in a soft play place, I needed leather, plastic and rope/net textures to complete the look. The full list of assets can be found in the references. What was a large determining factor in what texture to use was the inclusions of bump and normal maps, as lighting would play a large part in the look and feel of the game. There was an issue I came across with tiling, in which I needed to load textures multiple times over due to not being able to change their tiling once materialized, which heavily impacted how many textures looked.

Lighting

The lighting present throughout the game was varied and interesting! Due to the levels being indoors, I didn't get much chance for using hemisphere lighting, (or a skybox unfortunately), which led to my focus on spotlight and point lights. In the first scene, the point lights atop give a sense of security in being able to see quite clearly. They are supposed to feel artificial and industrial, akin to a warehouse. When the room goes dark, the spotlight from the torch then allows for players to see just enough to be able to move back to the start, but not enough to feel as comfortable. This especially works well in the second level in the dark foggy corridors and passageways, only illuminating as far as you can step. The second stage has a red spotlight over Harold too, for the added flair and intimidation factor.

Design

The game is quite unevenly split, with much more focus being put into the first level, less into the second due to the detached nature. The first level is set in the Happy House, and was planned to be poised with extra details: tables, images, extra toys to pick up and more. The softplay bit aimed to have a linear path for the player to follow to the end and back, with not enough time to add more sections to it. Building the soft play was the most difficult part due to all of the textures and requirements for the geometry to come together. However it was nice building the plastic walls with the small hexagonal bolts using prism geometry. The whole aim of the first level is to put you on edge while Harold stalks you from afar. Level 2 is much more the foundations of a level, currently with no achievable end goal, as Harold hunts you through the

halls to jump scare you. Each of these take the builder approach to making the level, by creating many functions to easily build the objects we need.

Limitations

The key limitation of the game is that it isn't finished. The set up of the controls, the flashlight pick up implementation, the physics for static and dynamic objects, and Harold's creepiness are all there, but with not enough time to properly implement them. Similarly level 2 is lacking because of no true way to win. The game provides a strong template and system to create more from it, yet is somewhat barebones itself. Amongst small things like extra textures, lore, models, the body of Harold, the game also could have been used with some of its glitches and frame drops ironed out. Its biggest limitation being, how it struggles to run on some devices.

Conclusion

Creating this game was a huge amount of fun, and it was great to make something feel creepy and off putting, while also working in lore ideas, and cool designs with lighting. It is far from perfect, and suffers from being more a proof of concept than a game, but it still holds value from its energy alone.

References

Code sources:

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